

1. Treasury risk from first principles

A bank transforms the timing, liquidity and risk characteristics of money. Depositors and wholesale creditors provide funding. The bank invests that funding in loans, securities, cash and other assets. Treasury manages the residual mismatches created by the business model.

The accounting identity is:

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

The identity does not tell us whether cash is available at the moment a payment is due. A bank can have positive equity and still face a payment shortfall because assets may be illiquid, pledged, legally trapped or not yet settled.

Liquidity and solvency

Dimension	Question	Typical measures
Solvency	Can losses be absorbed while liabilities remain covered?	Capital ratios, equity, loss-absorbing capacity
Liquidity	Can obligations be met when due without unacceptable loss?	Cash ladders, survival horizon, LCR, NSFR, intraday headroom
Funding	How reliable and diversified are the liabilities used to finance assets?	Maturity ladder, concentration, deposit behaviour, market access

Worked example

A bank has loans of 700, securities of 100, cash of 20, deposits of 650 and equity of 70. Assets are 820. Deposits plus equity provide 720, leaving a 100 wholesale funding requirement. If loans yield 6%, securities 4%, cash 3%, deposits cost 2% and wholesale funding costs 4.5%, annual NII is approximately $42 + 4 + 0.6 - 13 - 4.5 = 29.1$.

The funding gap and the earnings result answer different questions. Treasury must manage both.

2. Cash-flow timing, forecasting and survival

Liquidity is fundamentally a dated cash-flow problem. Forecasting should therefore be performed by value date, currency, legal entity and material settlement channel. Aggregating all inflows and outflows into a monthly total can conceal an earlier failure.

Core cash ladder

$$\text{Cash}_t = \text{Cash}_0 + \sum_{u \leq t} (\text{Inflows}_u - \text{Outflows}_u)$$

The survival horizon is the first time at which cumulative usable resources are insufficient to meet contractual and stressed obligations. Forecast error analysis should compare expected flows with actual settlements and assign causes: missing transactions, behavioural deviations, timing errors, wrong entity/currency tags and operational delays.

Worked example

Opening cash is 50. Daily expected receipts are 4 and payments are 5. Base daily cash use is 1. At day 30, cash is 20. Under a 30% receipt shortfall, receipts fall to 2.8 and daily cash use becomes 2.2. The first shortfall occurs after about 22.7 days. A small change in receipt reliability materially changes the survival horizon.

Implementation controls

- Reconcile opening cash to controlled ledger and settlement sources.
- Map flows to contractual and behavioural dates separately.
- Track forecast error by source and owner.
- Maintain currency/entity views; consolidated liquidity is not automatically transferable.
- Record management actions outside the passive cash ladder so they are not double-counted.